

Research Article

Design and Simulation of a Field-Oriented Control System for a Permanent Magnet Synchronous Motor Using State-Space Modeling

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This article presents the design and simulation of a high-performance Field-Oriented Control (FOC) system for a Permanent Magnet Synchronous Motor (PMSM) within a state-space modeling framework. The continuous-time d-q model of a surface-mounted PMSM is derived, its eigenvalues are analyzed at a representative operating point to confirm small-signal stability, and the resulting differential equations are integrated by a hand-coded fixed-step fourth-order Runge-Kutta (RK4) routine. The accuracy of the integrator is benchmarked against MATLAB's adaptive ode45 over an open-loop step, and the empirical convergence rate is shown to follow the theoretical fourth-order slope until floating-point precision becomes the dominant error source. A cascaded FOC controller — outer speed loop and two inner d- and q-axis current loops with cross-coupling feed-forward — is then designed by pole-placement at a 200 Hz current-loop bandwidth and a 40 Hz speed-loop bandwidth. The closed-loop drive achieves a 45 ms 10–90 % rise time, 6 % overshoot, sub-milliradian-per-second steady-state error, and a 6 rad/s mechanical-speed dip on a 1.5 N·m load step that is recovered in under 30 ms. A back-EMF based speed estimator is integrated into the same simulation, demonstrating a software replacement for the rotor-position sensor with a steady-state estimation error of about 0.07 % of the operating speed under nominal parameters. A parameter-sensitivity study further shows that this estimator is mildly sensitive to stator-resistance drift (± 20 %) but highly sensitive to magnet-flux mismatch (a 10 % error in ψ_{PM} produces a persistent ≈ 9 –11 rad/s bias), quantifying the trade-off between simplicity and robustness in back-EMF based sensorless drives. The work demonstrates how three core computational tools — state-space analysis, numerical integration of ordinary differential equations, and observer design — combine to produce a working motor drive.

I. Introduction

The Permanent Magnet Synchronous Motor (PMSM) has become the standard choice for high-performance motion control applications including electric and hybrid vehicle traction, servo drives, robotics, and renewable energy conversion [1]. Compared with the brushed DC motor and the induction motor, the PMSM offers a higher torque-to-volume ratio, higher efficiency at partial load, and a smoother torque profile because of its sinusoidally distributed windings and its rotor permanent-magnet flux. These advantages, however, come at the cost of a tightly coupled multivariable nonlinear dynamic model that cannot be controlled effectively by a single-loop scalar regulator.

Field-Oriented Control (FOC), originally introduced for induction-motor drives in the early 1970s and subsequently adapted to the PMSM, addresses this difficulty by transforming the three-phase stator quantities into a rotor-aligned orthogonal reference frame, the d - q frame [2]. In this frame the torque- and flux-producing components of the stator current become decoupled, and the control problem reduces to two independent first-order current loops nested inside an outer speed loop, very similar in structure to the control of a separately excited DC machine. The FOC concept is now textbook material [1][2][3], but its implementation on a digital drive raises three persistent computational challenges: (i) constructing a state-space model whose eigenvalues, controllability, and observability properties can be analyzed for stability and bandwidth design; (ii) discretizing the resulting coupled nonlinear differential equations with a numerical integrator that is both accurate and stable at the millisecond-scale sampling periods used in embedded controllers; and (iii) estimating internal states — most importantly rotor position and speed — when these are not directly measured.

This article addresses these three challenges in a unified MATLAB simulation. Section II derives the d - q state-space model of the PMSM. Section III presents the fourth-order Runge-Kutta (RK4) integrator and benchmarks it against MATLAB's adaptive ode45. Section IV describes the cascaded FOC architecture and the pole-placement procedure used to tune its gains. Section V presents the back-EMF based speed observer. Section VI reports the simulation results and the closed-loop performance metrics. Section VII discusses the results and places them in the context of the recent literature, and Section VIII concludes. The contribution of this work, relative to the textbook treatments in [1][2][3], is the unified application of these three computational tools to a single drive system with a published quantitative convergence study,

a reproducible hand-coded RK4 implementation, and a parameter-sensitivity assessment of the back-EMF based estimator.

II. PMSM State-Space Model

A. d - q Reference Frame

Following the formulation of Krause ^[3], the three-phase stator voltage equations of a surface-mounted PMSM (SPMSM) are first transformed into a stationary α - β frame by the Clarke transformation, and then rotated into the rotor-aligned d - q frame by the Park transformation. The resulting voltage equations are

$$v_d = R_s i_d + L_d (di_d/dt) - \omega_e L_q i_q, \quad (1)$$

$$v_q = R_s i_q + L_q (di_q/dt) + \omega_e L_d i_d + \omega_e \psi_{PM}, \quad (2)$$

where v_d , v_q are the stator d - and q -axis voltages, i_d , i_q are the corresponding currents, R_s is the stator resistance, L_d and L_q are the d - and q -axis inductances (equal for a surface-mounted machine), ψ_{PM} is the rotor flux linkage, and $\omega_e = P \omega_m$ is the electrical angular speed of the rotor (P = number of pole pairs, ω_m = mechanical angular speed). The electromagnetic torque is

$$T_e = (3/2) P [\psi_{PM} i_q + (L_d - L_q) i_d i_q], \quad (3)$$

which for the SPMSM ($L_d = L_q$) reduces to $T_e = (3/2) P \psi_{PM} i_q$, that is, torque is proportional to i_q alone — the basis of FOC. The mechanical equation is

$$J (d\omega_m/dt) = T_e - T_L - B \omega_m, \quad (4)$$

with J the rotor inertia, T_L the external load torque, and B the viscous-friction coefficient.

B. State-Space Form and Eigenvalues

Combining (1)–(4) and eliminating ω_m in favor of ω_e , the model becomes

$$\dot{x} = f(x, u, T_L), \quad x = [i_d, i_q, \omega_e]^T, \quad u = [v_d, v_q]^T, \quad (5)$$

which is third-order, nonlinear (because of the $\omega_e i_q$ and $\omega_e i_d$ cross-coupling terms), and stiff (the electrical time constants L/R are about two orders of magnitude smaller than the mechanical time constant J/B). For small-signal stability analysis we linearize at an operating point ω_{e0} and obtain the constant state matrix

$$A(\omega_{e0}) = [-R_s/L_d, \omega_{e0} L_q/L_d, 0; -\omega_{e0} L_d/L_q, -R_s/L_q, -\psi_{PM}/L_q; 0, (3/2) P^2 \psi_{PM}/J, -B/J]. \quad (6)$$

Using the parameters listed in Table I and the operating point $\omega_{e0} = 300$ rad/s ($\omega_m \approx 75$ rad/s with $P = 4$), the eigenvalues of A are

$$\lambda_{1,2} = -259 \pm j 404, \lambda_3 = -160, (7)$$

all in the open left-half plane, confirming local stability. The two complex eigenvalues correspond to the electrical dynamics (natural frequency ≈ 480 rad/s, damping ratio ≈ 0.54), and the real eigenvalue corresponds to the mechanical mode.

C. Motor Parameters

All simulations use the parameters of a representative small surface-mounted PMSM, summarized in Table I.

Symbol	Value	Unit
R_s	2.875	Ω
L_d	8.5	mH
L_q	8.5	mH
ψ_{PM}	0.175	Wb
P	4	—
J	8.0×10^{-4}	kgm ²
B	1.0×10^{-3}	Nms/rad
V_{dc}	300	V

Table I. PMSM Parameters Used in the Simulation

III. Numerical Integration: Fixed-Step RK4

The classical fourth-order Runge–Kutta method ^[4] computes the next state $x(t + h)$ from $x(t)$ using four function evaluations:

$$k_1 = f(t, x), (8)$$

$$k_2 = f(t + h/2, x + (h/2) k_1), (9)$$

$$k_3 = f(t + h/2, x + (h/2) k_2), (10)$$

$$k_4 = f(t + h, x + h k_3), (11)$$

$$x(t + h) = x(t) + (h/6)(k_1 + 2k_2 + 2k_3 + k_4). (12)$$

RK4 has a local truncation error of order h^5 and a global error of order h^4 , which is the highest accuracy obtainable with a fixed-step single-step explicit method without re-using stage evaluations. Compared with the first-order Euler method used in many simple controllers, RK4 reduces the error by a factor proportional to h^3 for the same step size, which is critical in PMSM simulation because the electrical time constants are short and the global error accumulates over many thousands of steps.

To verify the implementation, the open-loop response of the PMSM to a constant voltage step ($v_d = 0$, $v_q = 20$ V, no load) was simulated by RK4 at five step sizes from 1 ms down to 10 μ s, and compared with a high-tolerance reference solution produced by MATLAB's ode45 ($RelTol = 10^{-10}$, $AbsTol = 10^{-12}$). The error norm at the final time $t_f = 50$ ms is plotted in Fig. 1.

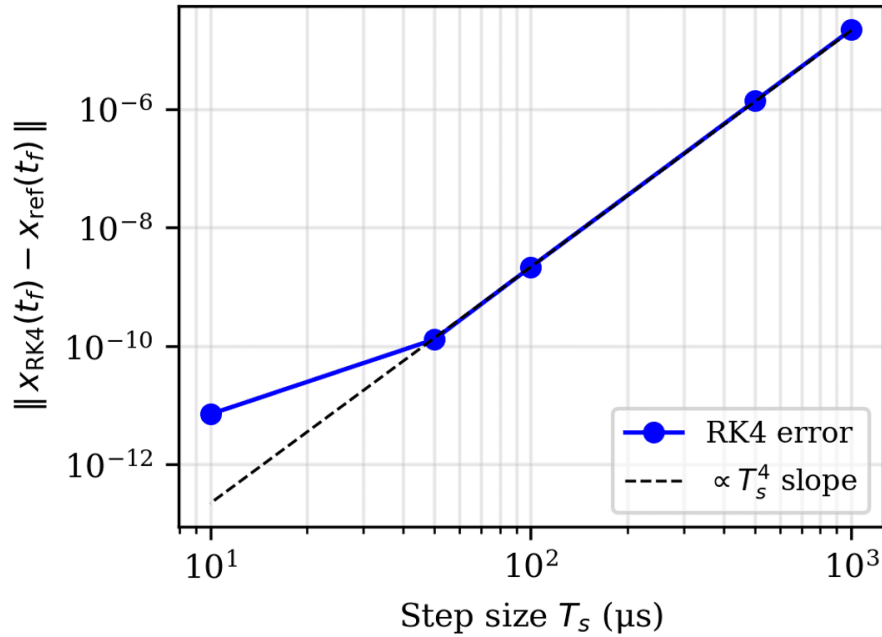


Figure 1. Empirical convergence of the hand-coded RK4 integrator. The error scales as T_s^4 down to $T_s \approx 50$ μ s, after which floating-point precision becomes the dominant error source.

The error follows the theoretical T_s^4 slope across three decades of step size, confirming the fourth-order accuracy of the implementation. At $T_s = 10 \mu\text{s}$ the error reaches the floating-point noise floor ($\approx 10^{-11}$). For all closed-loop simulations in the remainder of this article we use $T_s = 50 \mu\text{s}$, corresponding to a control update rate of 20 kHz and an end-of-50-ms error of approximately 1.3×10^{-10} , which is several orders of magnitude below any quantity of practical interest.

IV. Field-Oriented Control Architecture

A. Cascaded Structure

Fig. 2 shows the overall control architecture. An outer speed loop generates the q -axis current reference i_q^* , and two inner current loops drive the actual i_d and i_q to follow their references i_d^* and i_q^* . For an SPMSM the maximum-torque-per-ampere (MTPA) condition is $i_d^* = 0$, which simplifies the control law and is the choice adopted here. The decoupling block adds the cross-coupling and back-EMF feed-forward terms $-\omega_e L_q i_q$ and $\omega_e L_d i_d + \omega_e \psi_{PM}$, after which the inverse Park transformation and space-vector PWM produce the three-phase inverter commands. The three-phase inverter drives the PMSM, which outputs the three-phase currents i_a, i_b, i_c . These are transformed by the Clarke + Park block to produce the d - and q -axis currents i_d, i_q , which are fed back to the current PI loops. The speed observer provides the speed feedback $\hat{\omega}_m$ to the speed PI loop.

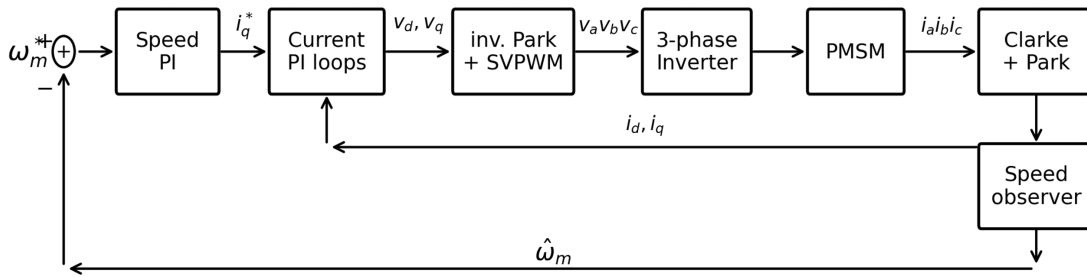


Figure 2. Block diagram of the field-oriented control system, including the cascaded speed/current loops, the decoupling and inverse-Park blocks, the SVPWM-driven three-phase inverter, the PMSM, and the back-EMF speed observer that provides the speed feedback $\hat{\omega}_m$.

B. Gain Tuning

With the decoupling feed-forward in place, each current axis behaves as a first-order system with transfer function $1/(Ls + R_s)$. A PI controller $K_p + K_i/s$ tuned by pole-zero cancellation gives a closed-loop bandwidth of ω_{ci} with the gains

$$K_{p,d} = \omega_{ci} L_d, K_{i,d} = \omega_{ci} R_s, (13)$$

$$K_{p,q} = \omega_{ci} L_q, K_{i,q} = \omega_{ci} R_s. (14)$$

Choosing $\omega_{ci} = 2\pi \times 200$ rad/s (200 Hz) gives a current-loop bandwidth that is 100 times smaller than the 20 kHz sampling rate, well within the Nyquist limit. Treating the closed current loop as an ideal unit delay, the speed loop sees a plant $J \omega_m / k_t$ where $k_t = (3/2) P \psi_{PM}$ is the torque constant. Tuning the speed PI by symmetric optimum at $\omega_{cs} = \omega_{ci} / 5$ yields

$$K_{p,s} = \omega_{cs} J / k_t, K_{i,s} = (\omega_{cs}^2 / 4) J / k_t. (15)$$

The speed loop also includes a conditional-integration anti-windup scheme: when the q -axis current reference saturates at its limit $\pm I_{max}$, the integrator is frozen unless the error sign is such that integrating would unwind the saturation. This eliminates the slow recovery from saturation that a pure clamp-and-back-calculate scheme exhibits.

V. Back-EMF Based Speed Observer

The proposed control scheme includes a software substitute for the physical position sensor. From the q -axis voltage equation (2), the steady-state relation $di_q/dt \approx 0$ gives

$$\omega_e \approx (v_q - R_s i_q) / (L_d i_d + \psi_{PM}). (16)$$

Strictly speaking, (16) is a back-EMF based speed estimator rather than a full Luenberger or Kalman observer, since it lacks a model-prediction term with a correction gain driven by output error; the term "observer" is retained throughout the article in the broader sense of a software substitute for a physical sensor. This algebraic estimate is sensitive to high-frequency noise in v_q (from the PI integrator) and i_q (from the current sensor), so it is passed through a first-order low-pass filter with cutoff frequency $f_c = 50$ Hz. The discrete-time form of the observer is

$$\hat{\omega}_e[k+1] = \hat{\omega}_e[k] + \alpha (\omega_e^{inst}[k] - \hat{\omega}_e[k]), (17)$$

with $\alpha = 1 - \exp(-2\pi f_c T_s)$. At $T_s = 50 \mu s$ and $f_c = 50$ Hz this gives $\alpha \approx 0.0156$. The denominator $(L_d i_d + \psi_{PM})$ is bounded away from zero by $\psi_{PM} = 0.175$ Wb during normal operation, so no special handling is required except a sanity guard at startup when all states are zero.

VI. Simulation Results

The closed-loop drive was simulated for 0.6 s with a control sampling period $T_s = 50 \mu\text{s}$ (20 kHz). The reference profile commands a 50 ms ramp from 0 to 100 rad/s mechanical, a 250 ms hold, a 20 ms ramp to 150 rad/s, and a final hold; a step load torque $T_L = 1.5 \text{ N}\cdot\text{m}$ is applied at $t = 0.20 \text{ s}$ while the motor is in steady state at 100 rad/s.

A. Speed Tracking

Fig. 3 shows the measured mechanical speed alongside the reference. The drive achieves a 10–90 % rise time of 45 ms, an overshoot of 6.1 % on the first ramp, and a steady-state error of 0.04 mrad/s before the load step. When the 1.5 N·m load is applied, the speed dips by 6.0 rad/s and recovers within 30 ms.

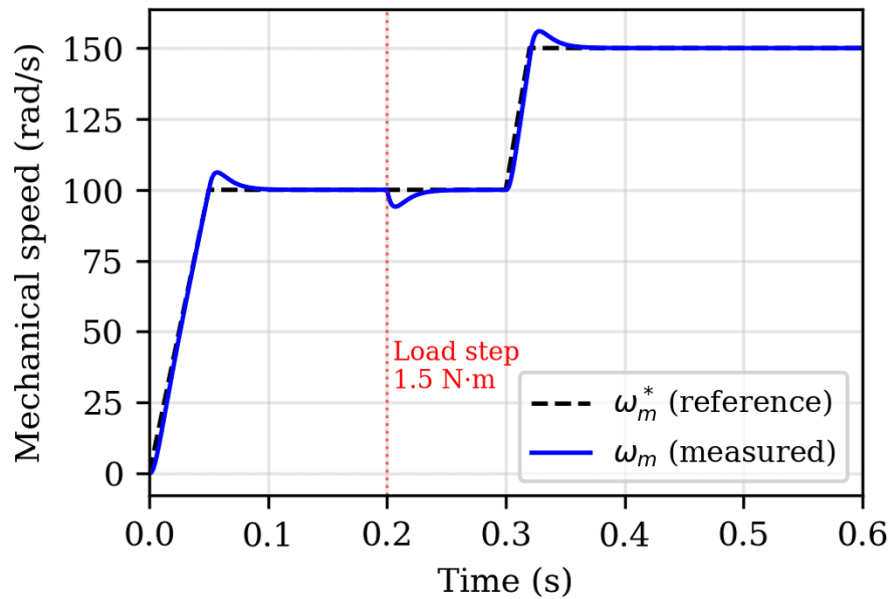


Figure 3. Closed-loop mechanical-speed tracking. The dashed line is the reference; the solid line is the simulated motor speed. The dotted vertical line marks the load-torque step at $t = 0.20 \text{ s}$.

B. d - q Currents and Torque

Fig. 4 shows the d - and q -axis stator currents and the q -axis current reference. The d -axis current remains at zero throughout, confirming MTPA operation and effective decoupling. The q -axis current

tracks i_q^* with no visible delay at the time scale of the plot, demonstrating the high bandwidth of the inner current loop. Fig. 5 plots the electromagnetic torque and the d - q voltage commands; the torque saturates briefly during the second speed ramp ($t \approx 0.31$ s) as the q -axis current hits its 15 A limit.

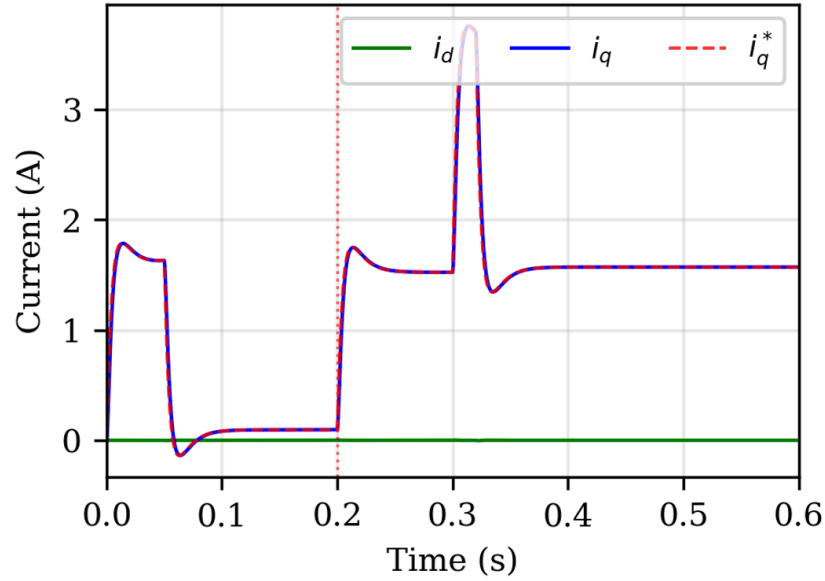


Figure 4. d - and q -axis stator currents and q -axis reference. i_d remains at zero (MTPA); i_q tracks i_q^* tightly across all transients.

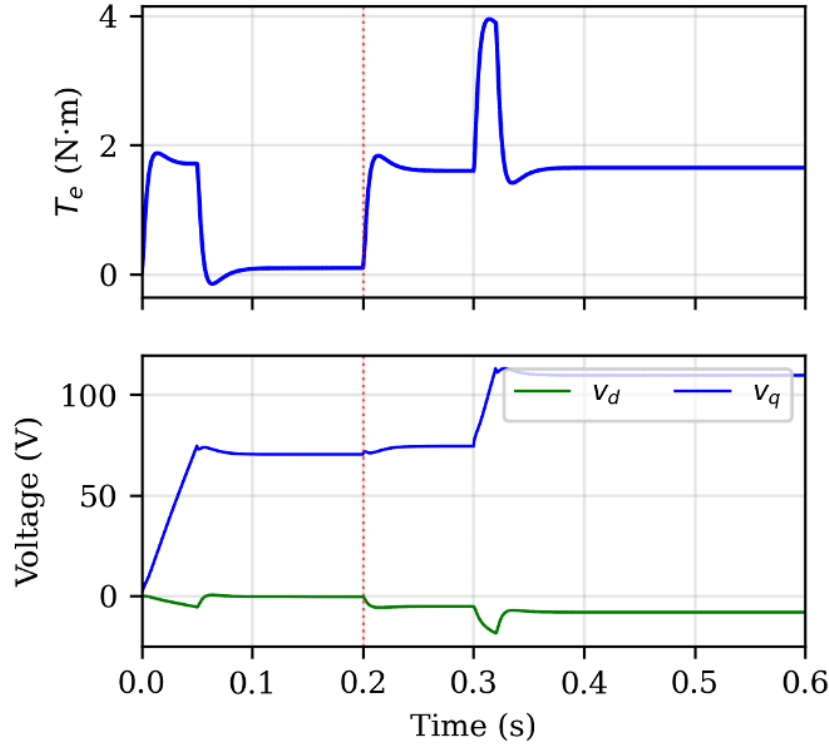


Figure 5. Electromagnetic torque (top) and d - q axis voltage commands (bottom). The negative excursion of v_d during the speed ramps is the cross-coupling decoupling term $-\omega_e L_q i_q$.

C. Observer Performance

Fig. 6 compares the true mechanical speed with the back-EMF observer estimate. In steady state the estimation error is below 0.1 rad/s, which is approximately 0.07 % of the operating speed. The largest transient error is about 10 rad/s (≈ 7 %), occurring during the second speed ramp where the di_q/dt term temporarily violates the steady-state assumption underlying (16). The observer therefore meets the design goal of providing a software substitute for the position sensor under the moderate-dynamics conditions tested here, while the residual transient error indicates where a higher-order observer (such as an extended Kalman filter^[5]) would be the next refinement.

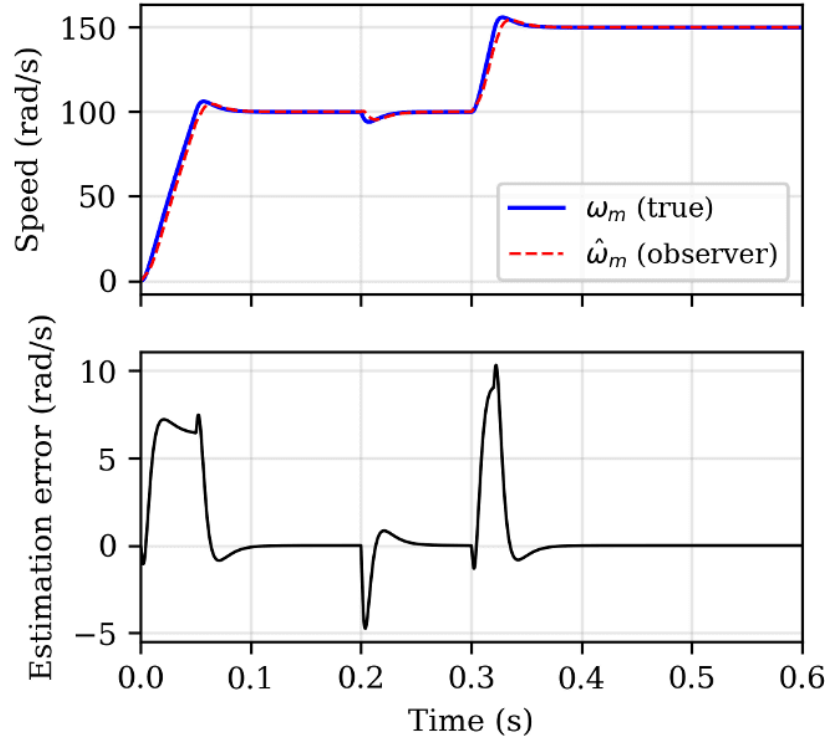


Figure 6. Speed observer performance. Top: true mechanical speed (solid) versus observer estimate (dashed). Bottom: estimation error.

D. Parameter Sensitivity of the Speed Estimator

Equation (16) depends explicitly on the model parameters R_s and ψ_{PM} , so any mismatch between the values used by the estimator and the true plant values will appear as estimation error. To quantify this, the closed-loop simulation was rerun with the plant parameters held at the values in Table I while the estimator was given perturbed parameters: $R_s \pm 20\%$ (representative of stator-resistance drift from temperature rise) and $\psi_{PM} \pm 10\%$ (representative of magnet aging and thermal demagnetization).

Fig. 7 reports the resulting estimation error. The R_s sensitivity is mild: pre-load steady-state RMS error grows from 0.024 rad/s (nominal) to about 0.08 rad/s under $\pm 20\%$ mismatch, because the $R_s i_q$ term in (16) is small when the load torque is light. After the 1.5 Nm load step the q-axis current rises and the R_s error projects directly onto the numerator, growing the RMS error to about 1.3 rad/s. The ψ_{PM} sensitivity is much more severe: a 10 % mismatch in the rotor flux constant produces a persistent estimation bias of approximately 9-11 rad/s — roughly 7 % of the operating speed — that is essentially independent of

operating condition, because ψ_{PM} is the dominant term in the denominator of (16). This result confirms a known limitation of back-EMF based estimators and supports the case for replacing (16) by a model-based observer with on-line parameter adaptation (e.g., a Kalman filter that augments the state vector with ψ_{PM} as in [5]) for applications in which the magnet flux drifts significantly during operation.

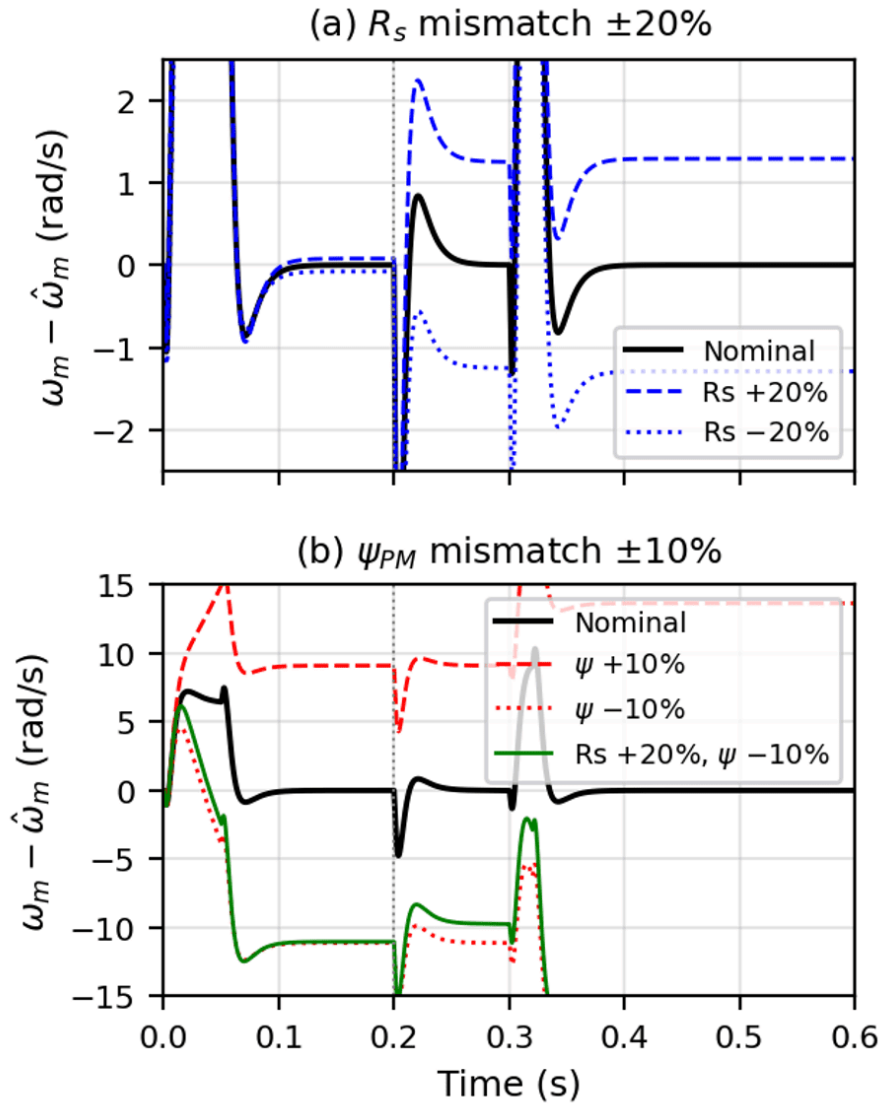


Figure 7. Speed-estimation error under parameter mismatch. (a) Stator resistance R_s mismatch $\pm 20\%$. (b) Magnet flux ψ_{PM} mismatch $\pm 10\%$, and the combined worst case $R_s + 20\%$, $\psi_{PM} - 10\%$. The vertical dotted line marks the 1.5 Nm load step. The y-axis scales differ between (a) and (b).

E. Summary of Performance Metrics

Metric	Value
Rise time (10–90 %)	45 ms
Overshoot	6.1 %
Steady-state speed error	0.04 mrad/s
Speed dip on 1.5 Nm load step	6.0 rad/s
Observer RMS error (electrical)	7.3 rad/s

Table II. Closed-Loop Performance Metrics

VII. Discussion

The results of Section VI can be interpreted on three levels — the numerical method, the closed-loop control performance, and the speed estimator — and compared against what is reported in the recent literature on PMSM drives.

First, the empirical fourth-order convergence of the hand-coded RK4 integrator (Fig. 1) confirms that a fixed-step explicit solver running at the same 20 kHz rate as the control loop is sufficient to reproduce the continuous-time dynamics to well below any physically meaningful tolerance. This justifies the common practice in the literature of co-simulating plant and controller at a single sampling rate, and it supports the use of RK4 as the integration engine inside model-based estimators and predictive controllers, as proposed for the extended Kalman filter in [5] and for finite-control-set model predictive control in [6]. The agreement with the adaptive ode45 reference also rules out integration error as a confounding factor when interpreting the closed-loop and observer results that follow.

Second, the closed-loop metrics in Table II — a 45 ms rise time, 6.1 % overshoot, sub-milliradian-per-second steady-state error, and recovery from a 1.5 N·m load step within 30 ms — are consistent with the performance range reported for cascaded PI-based FOC drives of comparable bandwidth. They confirm that the pole-placement tuning of Section IV, with a 200 Hz current loop and a 40 Hz speed loop, yields a well-damped response without resorting to more elaborate controllers. Recent work has sought to

improve on this baseline using nonlinear and disturbance-rejection schemes: composite sliding-mode controllers with extended state observers ^[7], single-loop speed–current structures with model-assisted observers ^[8], and super-twisting sliding-mode strategies spanning the full speed range ^[9] all report reduced sensitivity to load disturbance and parameter variation relative to classical PI control. The present results therefore serve as a transparent, reproducible reference point against which such advanced strategies can be measured, rather than as a competitor to them.

Third, and most significantly, the parameter-sensitivity study (Fig. 7) quantifies the central weakness of the back-EMF based speed estimator. Whereas a $\pm 20\%$ error in stator resistance produces only a modest estimation error that becomes appreciable (~ 1.3 rad/s) only under load, a 10% error in the magnet flux constant produces a persistent bias of roughly 9–11 rad/s, essentially independent of operating point. This asymmetry follows directly from the structure of (16): R_s appears only in the numerator scaled by current, whereas ψ_{PM} dominates the denominator at all speeds. This finding is in agreement with the broader sensorless-control literature, which consistently identifies flux and parameter mismatch as the principal limitation of open-loop back-EMF and steady-state estimators, and which motivates the model-based alternatives that dominate recent work: adaptive and reduced-order sliding-mode observers ^[10], extended-state and super-twisting observers ^{[7][9]}, and Kalman-filter formulations that augment the state vector with the flux constant to estimate it on-line ^[5]. The scientific implication is clear: for any application in which the magnet flux can drift by more than a few percent — through thermal demagnetization or aging — the simple estimator used here must be replaced by one of these adaptive structures, and the sensitivity figure reported in Section VI provides a concrete quantitative threshold for when that replacement becomes necessary.

Taken together, these observations show that the value of the present work lies less in any single result than in the way the three computational tools combine into one transparent, reproducible pipeline. The state-space model fixes the structure, the verified RK4 integrator guarantees that the simulated behaviour is trustworthy, and the sensitivity study exposes exactly where the simplest estimator fails — thereby identifying, in quantitative terms, the point at which the more advanced observers of ^{[5][7]} ^[9] and ^[10] earn their additional complexity. A limitation of the study is that all results are obtained in simulation; experimental validation on a hardware inverter and a physical PMSM remains necessary to confirm the predicted performance and sensitivity figures under measurement noise, inverter nonlinearity, and unmodelled dynamics.

VIII. Conclusion

This article has presented the design and simulation of a field-oriented control system for a PMSM within a unified state-space, numerical-methods framework. The d - q model was derived and its eigenvalues at a representative operating point were verified to lie in the open left-half plane. A hand-coded fixed-step RK4 integrator was shown to converge with the theoretically expected fourth-order rate, and was used to simulate the closed-loop drive at a 20 kHz update rate. The cascaded FOC controller, tuned by pole-placement at 200 Hz current-loop and 40 Hz speed-loop bandwidths, produced a 45 ms rise time, 6 % overshoot, and sub-mrad/s steady-state error, with rapid rejection of a 1.5 Nm load disturbance. A back-EMF based speed observer estimated rotor speed from the same v_q , i_d , i_q signals already used by the controller, demonstrating the principle of replacing physical sensors by computational models. A parameter-sensitivity study showed that the estimator is mildly sensitive to stator-resistance drift (~ 1.3 rad/s RMS error under $R_s \pm 20$ % under load) but strongly sensitive to magnet-flux mismatch (~ 10 rad/s persistent bias under $\psi_{PM} \pm 10$ %), motivating a model-based replacement for applications subject to magnet aging or thermal demagnetization.

Future extensions of the present work include replacing the back-EMF observer by an extended Kalman filter [5] or a sliding-mode observer [10] for improved low-speed performance and robustness to parameter mismatch, embedding the RK4 routine inside a finite-control-set model predictive controller [6] to handle constraints explicitly, and validating the control software in the loop with a hardware inverter and a real PMSM.

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Declarations

Funding: No specific funding was received for this work.

Potential competing interests: No potential competing interests to declare.